|INCH-POUND|

MIL-B-80160C
16 March 1992
SUPERSEDING
MIL-B-80160B
17 June 1982

MILITARY SPECIFICATION

BENDING PRESSES, RAM TYPE, HYDRAULICALLY DRIVEN, PIPE, TUBING AND STRUCTURAL SHAPES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers ram type bending presses for forming radius bends on metal pipe, tubing and structural shapes by wrapping the workpiece around a ram or punch die, with equipment and accessories specified herein.

1.2 Classification. The presses shall be of the following sizes. The size to be furnished shall be as specified (see 3.5 and 6.2).

Size 1 - 3 tons
Size 2 - 5 tons
Size 3a - 14 tons
Size 4a - 25 tons
Size 5 - 30 tons
Size 6 - 40 tons

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, 2163 Airways Blvd. Memphis, TN 38114-5213, by using the self-addressed Standardization|Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U.S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402-0001.)

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(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contract activity.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B11.15 - Pipe, Tube, and Shape Bending Machines - Safety Requirements for the Construction, Care, and Use. (DOD adopted)

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3363.)

NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

NEMA MG 1 - Motors and Generators. (DOD adopted)

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, N.W., Washington, DC 20037-1580.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery. (DOD Adopted)

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

NATIONAL FLUID POWER ASSOCIATION, INC. (NFPA)

NFPA/JIC T2.24.1 - Hydraulic fluid power - Systems standard for industrial machinery.

(Application for copies should be addressed to the National Fluid Power Association Inc., 3333 N. Mayfair Road, Milwaukee, WI 53222-3291.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), one complete machine shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

3.2 Design. The machine shall be new (not a prototype) and one of the manufacturer's current production models capable of operations in accordance with the requirements herein. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be readily accessible for adjustment, replacement, or repair. The machine shall be designed to form true radius bends, crush bends, and collapse bends in tubing, pipe, and structural shapes, by a compression ram or punch die pressure against opposing rotating wing dies, to wrap the workpiece around the ram die.

3.2.1 Measuring and indicating device calibration. Measuring and indicating devices shall be graduated in inch-pound (see 6.5.1), metric (see 6.5.2), or dual (inch-pound and metric) units as specified by the procuring activity (see 6.2). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.1.1 Dials. All dials used to indicate angle of workpiece bend shall be graduated in increments not larger than 1/2 degree. Dial and handwheel circumferences shall be permanently and legibly engraved or etched on a nonglare background with graduations that can be read from the operator's normal position.

3.2.1.2 Digital readout and control system. When specified (see 6.2), a digital readout system shall be installed on the machine. The system shall consist of an electronic display and control for the angle of bending with related components, mounting hardware, and electrical equipment necessary to ensure complete operation of the system. The system shall provide a readily visible digital display over the full working range of the machine on which installed.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials to the maximum extent possible provided such materials will not jeopardize the intended use, performance, or design life of the machine. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used, rebuilt, or remanufactured end products are allowed under this specification.

3.2.3 Energy efficiency. The machine and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency in accordance with the developments available within the industry.

3.2.4 Controls. All mechanical, hydraulic, electrical, and pneumatic operating controls shall be located convenient to the operator's work station. The controls shall provide for automatic cycling of the machine and shall control the angle of the bend to plus or minus 1/2 degree of the selected angle.

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3.2.5 Safety and health requirements. All parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall conform with all requirements of OSHA 29 CFR 1910 and ANSI B11.15. If a conflict arises between OSHA 29 CFR 1910 and ANSI B11.15, OSHA 29 CFR 1910 shall apply. Additional safety and health requirements shall be as specified (see 6.2).

3.2.5.1 Noise level. Unless otherwise specified (see 6.2), noise emitted by the machine when operating under a no load condition shall not exceed 84 db when measured on the A scale of a type 1 or type 2 sound level meter at slow response.

3.2.6 Environmental protection. During the manufacture, service, transportation, or storage of the machine and its components, the machine shall meet all applicable Environmental Protection Agency (EPA) and OSHA restrictions for materials classified as hazardous to the environment in effect on the date of the contract. The operation of the machine at its final location shall comply with all applicable EPA and OSHA regulations in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the operator's health. Specific environmental protection requirements exceeding the established EPA and OSHA regulations shall be as specified by the procuring activity (see 6.2).

3.2.7 Lubrication. Means shall be provided to ensure lubrication for all moving parts. Recirculating lubrication systems shall include a cleanable or replaceable filter. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.8 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. All parts shall be manufactured to such standards as will permit replacement or adjustment without modification of parts or machine.

3.3 Construction. The machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion or failure.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Fastening devices. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

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3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein (see 6.2), the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2), the paint color and application shall be in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Gears. All gears shall be constructed from a material suitable for the intended purpose. All drive gears shall have the hardness, surface finish, and toughness that will enable the gear train to transmit full rated torque of the drive motor without gear damage, failure, or premature deterioration and wear.

3.3.8 Electromagnetic interference control. When specified (see 6.2), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystem(s) class and the emission and susceptibility requirements shall be as specified.

3.4 Components. The machine shall consist of not less than a base and frame, ram, wing die assembly, hydraulic system, electrical requirements, and additional components necessary for the operation of the machine.

3.4.1 Base and frame. The machine base and frame shall be a casting of iron or a steel plate weldment, with machined ways for the ram. The frame shall provide for mounting the required components of the machine.

3.4.2 Ram. Unless otherwise specified (see 6.2), the ram shall be mounted on the machine in either a vertical or horizontal position, manufactured of steel and machined to mount the bending dies. The ram shall be mounted in sliding ways which are fitted with replaceable wear strips. The ram shall carry the punch die, which will form the contour and radius of the bend.

3.4.3 Wing die assembly. The wing die assembly shall provide for the "wrapping" or forming of the workpiece around the punch or form die. The wing die holders shall pivot on spindles, provide for the mounting of interchangeable dies, and for the application of controlled pressure by the fully controlled die assembly movement over the workpiece.

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3.4.4 Hydraulic system. The hydraulic system shall conform to or exceed the requirements of NFPA/JIC T2.24.1 and shall be complete, including all pumps, gauges, valves, piping, cylinders, and pressure controls. Overpressure protection shall be provided in the high pressure line to prevent damage to components. A filter system shall be provided to ensure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level, provisions for draining, and a cleanout access plate.

3.4.5 Electrical requirements. The machine shall be wired to conform to NFPA 79. Each machine shall be wired to draw all of its electrical power from a single circuit. Unless otherwise specified (see 6.2), each machine shall be designed to operate on 3-phase, 60-Hertz (Hz), dual 230/460-volts. Conversion from one voltage to the other shall be accomplished by the reconnection of existing components. The machine shall be initially wired as specified (see 6.2).

3.4.5.1 Motors. Motors shall meet the requirements of 3.4.5. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2), alternating current (AC) motors shall be designed to operate on 60-Hz. The motor enclosure shall meet the requirements for a drip-proof enclosure. All motors shall conform to the requirements of NEMA MG-1.

3.4.6 Worklight. When specified (see 6.2), a worklight shall be provided. The light shall have a protective shield, an on-off switch, and shall be adjustable to reflect maximum light on the work area. If a built in lighting system is provided, the adjustable light described above shall not be required.

3.4.7 Hour meter. When specified (see 6.2), the machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 9,999 hours in increments of not greater than 1 hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine.

3.5 Size and capacity. Unless otherwise specified (see 6.2), the size and capacity of the machine shall meet the requirements of table I for the size specified. Machine size and capacity shall be not less than the stated requirements of the size ordered. When a range is given, the required performance shall be from the stated minimum or less to the stated maximum or greater.

3.5.1 Physical size limitations. When specified (see 6.2.), the physical size of the machine shall not exceed the height, width, and length restrictions specified by the procuring activity to ensure the machine will fit in its future operating location.

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TABLE I. Sizes and capacities.

Required capacity, (mild steel)	Size 1 3 tons	Size 2 5 tons	Size 3a 14 tons	Size 4a 25 tons	Size 5 30 tons	Size 6 40 tons
Maximum capcity steel tubing						
One tube	1-1/8" OD x .065"		1-1/2" OD x .065"	2-1/4" OD x.065"	3" OD x .109"	3-1/2" OD x .125"
Two tubes (per stroke)		1" OD x .065"	1-1/8" OD x .065"	1-1/8"OD x .083"	1-3/4"OD x .125"	2" OD x .125"
Maximum angle*	140 3-1/2 CLR	100 3-1/2 CLR	150 5-1/2 CLR	165 6" CLR	165 10" CLR	165 10" CLR
Maximum CLR	4-1/2"	3-1/8"	3-15/16"	6"	10"	10"
Minimum CLR	2D	2D	2D	4"	4"	4-1/2"
Bends per hr. (approx.)	514	1800 (2 tubes)	600	800	800	600

* NOTE: Increasing the centerline radius above those listed shall reduce the maximum angle proportionally.

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3.6 Performance. The press shall be capable of forming ferrous and non-ferrous tubing, pipe, and structural shapes of mild steel with yield point of up to 40,000 psi, to any desired angle from 0 to maximum for the centerline radius as listed in table I for the size machine in the contract or purchase order. The press shall also be capable of forming true radius, crush, and collapse bending operations.

3.6.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2).

3.7 Standard equipment. All standard equipment normally provided with the manufacturer's commercial machine shall be furnished.

3.8 Optional and accessory equipment. Optional and accessory equipment shall be furnished as specified and fully described (see 6.2). The equipment provided shall be functional without requiring modification of the equipment or machine.

3.9 Repair parts. Required repair parts shall be furnished as specified (see 6.2).

3.10 Marking on instruments, control panels, charts, and plates. All words on the machine including all instruments, control panels, control units, charts, plates, and related attachments shall be in the English language. Characters shall be permanently marked in boldface on a contrasting background. All plates shall be corrosion resistant.

3.10.1 Lubrication plate or chart. When specified (see 6.2) a lubrication plate or chart shall be attached to each machine. If a chart is furnished it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets. Unless otherwise specified (see 6.2) the information provided on the chart or plate shall be as listed below.

Points of lubricant application
Type of lubricant
Servicing interval
SAE number or military or Federal specification for each lubricant (if applicable)

3.10.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2), the nameplate shall contain the information listed below.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Amp rating of largest motor
Short circuit/over-current protection rating
Contract number or order number
National stock number or plant equipment code
Date of manufacture

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3.10.3 Hydraulic fluid requirements plate. When specified (see 6.2), a hydraulic fluid requirements plate shall be attached to the machine at the hydraulic reservoir fill point. Unless otherwise specified (see 6.2), the plate shall contain the following information:

Type of fluid (SAE number or military or Federal specification number)
Type of filter

3.11 Workmanship. Workmanship of the machine and its accessories shall meet all requirements specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in 4.7.1.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. The machine shall pass the examination and all tests to be accepted.

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4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each machine prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.6, the test(s) in 4.7, and the inspection in 4.8. The machine shall pass the examination, all tests, and the inspection to be accepted.

4.6 Examination. Each machine shall be examined to determine compliance with all requirements of this specification.

4.7 Tests.

4.7.1 Test conditions. All tests shall be performed in an indoor facility with ambient conditions of 41° to 104° Fahrenheit (F) and 20 percent to 95 percent relative humidity.

4.7.2 Operational test. Each machine shall be operated at no load for not less than 1/2 hour. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.7.3 Performance tests. The machine shall be set up to bend workpieces of maximum capacity and to the maximum degree of bend to form true radius, crush, and collapse bends for the machine size being tested. The machine shall then be operated to bend six workpieces, as specified in table I, not less than two feet long, to the same angle of bend without further adjustment. The six workpieces shall be examined to determine if the degree of bend is within 1/2 degree of the set up bend and is free from wrinkles or distortion. All performance tests shall be completed without making any adjustments to or performing any maintenance on the machine.

4.7.3.1 Additional performance tests. Performance tests required in addition to or in lieu of the requirements of 4.7.3 shall be as specified and fully described (see 6.2).

4.7.4 Noise level test. Noise level measurements shall be made to ensure conformance with 3.2.5.1. Unless otherwise specified (see 6.2), the measurements shall be made at the operators work station and each corner of the machine. Noise emitted by the machine when operating under a no load condition shall not exceed 84 db when measured on the A scale of a type 1 or type 2 sound level meter at slow response. The test shall be conducted with the machine operating at normal speeds.

4.7.5 Environmental protection certification of conformance. The con-contractor shall certify and maintain substantiating evidence that the product offered complies with all environmental protection restrictions as required in 3.3.6. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

4.7.6 Electromagnetic interference control test. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.8 using the procedures given in MIL-STD-462.

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4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-18058.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Machines produced to this specification are intended for use in bending pipe,tubing,and structural shapes in production shops where tolerances of plus or minus 1/2 degree are required.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Size of machine required (see 1.2 and 3.5).

c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

d. First article, if required (see 3.1).

e. System of units for measuring and indicating devices (inch-pound) (see 3.2.1).

f. Digital readout system, if required. Specify any special digital readout system features (see 3.2.1.2).

g. Additional safety and health requirements, if required (see 3.2.5).

h. Specify noise level requirements, if different (see 3.2.5.1).

i. Environmental protection requirements exceeding EPA/OSHA requirements (see 3.2.6).

j. Specify surface finish requirements, if different (see 3.3.3).

k. Painting, if different (see 3.3.5.).

l. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.8).

m. Specify ram requirements, if different (see 3.4.2).

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n. Voltage requirements, if different (see 3.4.5).

o. Initial voltage, as specified (see 3.4.5).

p. Motors, if different (see 3.4.5.1).

q. Worklight, if required (see 3.4.6).

r. Hour meter, if required (see 3.4.7).

s. Size and capacity, if different (see 3.5).

t. Specify physical size limitation, if required (see 3.5.1).

u. Additional performance features, if required (see 3.6.1).

v. Specify and fully describe optional equipment, if required (see 3.8).

w. Specify applicable repair parts, if required (see 3.9).

x. Lubrication plate, if required (see 3.10.1).

y. Lubrication plate information, if different (see 3.10.1).

z. Nameplate, if different (see 3.10.2).

aa. Hydraulic fluid requirements plate, if required (see 3.10.3).

bb. Hydraulic fluid requirements plate, if different (see 3.10.3).

cc. Inspection conditions, if different (see 4.3).

dd. First article inspection, if different (see 4.4).

ee. Quality conformance inspection, if different (4.5).

ff. Additional performance tests, if required (see 4.7.3.1).

gg. Noise level acceptance test, if different (see 4.7.4).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specification acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

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Reference Paragraph	DID Number	DID Title	Suggested Tailoring
N/A	DI-TMSS-80527	Commercial Off-The-Shelf (COTS) Manuals	Require all data be in the English Language
N/A	DI-TMSS-80528	Supplemental Data for Commercial Off-The-Shelf (COTS) Manuals	Require all data be in the English Language

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Definitions.

6.5.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined by the National Institute of Standards and Technology. Inch-pound units having the same names in other countries may differ in magnitude.

6.5.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le Systeme International d'Unites (SI)," of the International Bureau of Weights and Measures. These units are described in ASTM E 380 and IEEE 268 (MIL-STD-961).

6.6 Measurement system. In this specification, all measurements, dimensions, sizes, and capacities are given in inch-pound units. These measurements may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.

6.7 Cross-reference of classification. Classification changes from the previous issue of this specification are as follows:

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MIL-B-80160B	MIL-B-80160C
Size 3	Size 3 (deleted)
Size 4	Size 4 (deleted)
	Size 3a (new)
	Size 4a (new)

6.8 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

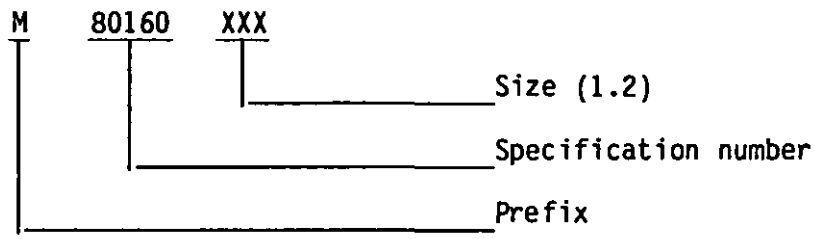
6.9 Safety and health requirements. OSHA 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new machine into its future environment.

6.10 Training. Training required by the procuring activity should be provided as specified in the contract.

6.11 Warranty. Warranty requirements should be as specified by the procuring activity in the contract.

6.12 Inspection location. The contractor should identify in his response to the solicitation the location where inspection and tests are to be performed.

6.13 Part or Identifying Number (PIN). The PINs to be used for machines acquired to this specification are created as follows:



6.14 Subject term Key word listing.

Bending
Forming
Machine
Metal
Press
Ram, forming

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Check the source to verify that this is the current version before use.

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activity:
Air Force - 84
DLA - GS

User activity:
Navy - YD

Preparing Activity:
DLA - IP

Project (3441-0156)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-B-80160C	91 03 16

3. DOCUMENT TITLE: BENDING PRESSES, RAM TYPE, HYDRAULICALLY DRIVEN, PIPE, TUBING AND STRUCTURAL SHAPES

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
Defense Industrial Plant Equipment Center ATTN: DIPEC-SSG	(1) Commercial (901)-775-6694	(2) AUTOVON 683-6694
c. ADDRESS (Include Zip Code) 2163 Airways Blvd. Memphis, TN 38114- 5213	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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